"Uncomplicated" capacitor

W22 (2022) — English translation

Part A. Charged plate in a capacitor.

In the middle between the identical uncharged metal plates of a flat capacitor with an area S and a distance between them $d \ll \sqrt{S}$ there is a dielectric plate of mass m of the same shape, charged uniformly over the surface with a charge $q > 0$. The capacitor plates were connected to a constant voltage source $\mathcal{E}$ with zero internal resistance. Then the plate is released. When moving, the capacitor plates remain motionless. Let us denote by x the deviation of the plate from the initial position, by q_1 the charge of the left plate, and by q_2 the charge of the right plate.

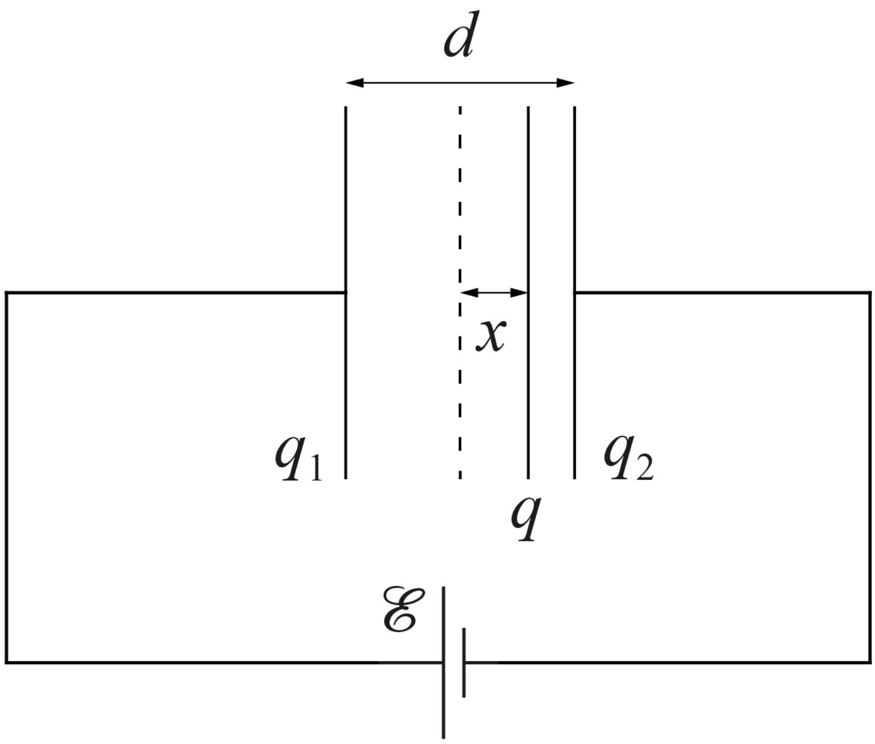

Figure 1: Figure 1.

A1	Express the charges of the capacitor plates $q_1(x)$ and $q_2(x)$ in terms of $\mathcal{E}$, S, d, q, x and physical constants.	**0.50**

A2	Express the force of electrostatic interaction $F(x)$ of the plate with the plates in terms of $\mathcal{E}$, S, d, q, x and physical constants.	**0.50**

It can be shown that the dependence of the displacement of the plate $x(t)$ on time has the following form

$$x(t) = Ae^{at} + Be^{-at} + C,$$

where a depends only on $\mathcal{E}$, S, d, q and m, while A, B and C are determined by the initial conditions.

A3	Express a, A, B and C in terms of $\mathcal{E}$, S, d, q, m and physical constants.	**1.00**

A4	Express the speed of the plate v^* at the moment of impact on the capacitor plate in terms of $\mathcal{E}$, S, d, q, m and physical constants.	**1.00**

Part B. Capacitor with Dielectric

A flat capacitor with plate area S and distance between them $d \ll \sqrt{S}$ is filled with a dielectric with dielectric constant ε and resistivity ρ. The dielectric and the plates have good electrical contact. Let the potential difference between the plates of the capacitor be V.

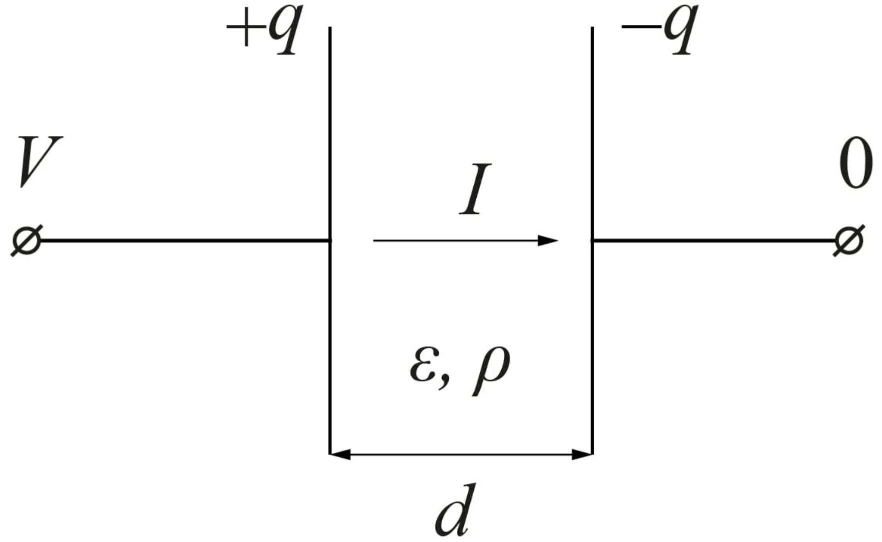

Figure 2: Figure 2.

B1	Find the charges q on the metal plates of the capacitor. Express your answer in terms of V, S, d, ε and physical constants.	**0.30**

B2	Find the current I flowing through the dielectric. Express your answer in terms of V, S, d, ρ and physical constants.	**0.30**

From the two previous points it follows that the circuit of such a capacitor in an electrical circuit is equivalent to some connection of a resistor with resistance R and a capacitor with capacitance C.

B3	Find R and C, and also indicate the type of connection of these elements (series or parallel).	**0.40**

Now the complex capacitor is connected to a constant voltage source $\mathcal{E}$, which has an internal resistance r. At the initial moment of time, the capacitor is not charged.

B4	Find the time dependence of the current $I(t)$ flowing through the source. Express your answer in terms of $\mathcal{E}$, r, R, C and t.	**2.00**

Part C. Two-Dielectric Capacitor

A flat capacitor with a distance between plates d is filled with two dielectrics. One with dielectric constant ε_1, resistivity ρ_1 and contact area with plates S_1, and the other with dielectric constant ε_2, resistivity ρ_2 and contact area with plates S_2. Both dielectrics have good electrical contact with the capacitor plates.

C1	Find the equivalent capacitance C_0 of this capacitor. Express your answer in terms of S_1, S_2, ε_1, ε_2, d and physical constants.	**0.40**

C2	Find the equivalent resistance R_0 of this capacitor. Express your answer in terms of S_1, S_2, ρ_1, ρ_2 and d.	**0.40**

This capacitor in series with the capacitor C is connected to a constant voltage source $\mathcal{E}$, which has zero internal resistance. While the key was open, the capacitors were not charged. The voltage across a complex capacitor is measured with an ideal voltmeter V.

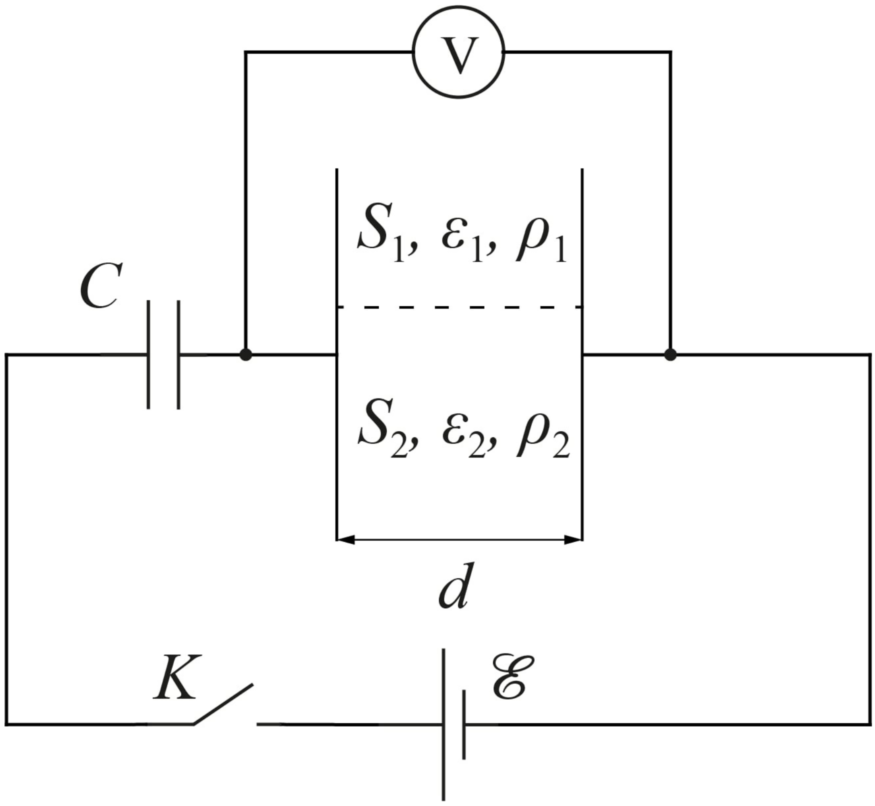

Figure 3: Figure 3.

C3	Find the charge $q_C(0)$ on the capacitor capacitance C immediately after closing the switch. Express your answer using $\mathcal{E}$, C and C_0.	**0.40**

C4	Find the dependence of the voltmeter readings $V(t)$ on time t after closing the key. Express your answer in terms of $\mathcal{E}$, C, C_0, R_0 and t.	**1.30**

C5	How much heat Q_{R_0} will be released at resistance R_0? Express your answer using $\mathcal{E}$, C and C_0.	**0.50**

Useful integrals:

$$\int \frac{dx}{ax+b} = \frac{1}{a}\ln(ax+b) + C, \qquad \int e^{ax}\,dx = \frac{e^{ax}}{a} + C.$$

Source: `https://pho.rs/p/271`